Manipulating antioxidant intake in asthma: a randomized controlled trial.

BACKGROUND: Antioxidant-rich diets are associated with reduced asthma prevalence in epidemiologic studies. We previously showed that short-term manipulation of antioxidant defenses leads to changes in asthma outcomes. OBJECTIVE: The objective was to investigate the effects of a high-antioxidant diet compared with those of a low-antioxidant diet, with or without lycopene supplementation, in asthma. DESIGN: Asthmatic adults ($n = 137$) were randomly assigned to a high-antioxidant diet (5 servings of vegetables and 2 servings of fruit daily; $n = 46$) or a low-antioxidant diet (≤ 2 servings of vegetables and 1 serving of fruit daily; $n = 91$) for 14 d and then commenced a parallel, randomized, controlled supplementation trial. Subjects who consumed the high-antioxidant diet received placebo. Subjects who consumed the low-antioxidant diet received placebo or tomato extract (45 mg lycopene/d). The intervention continued until week 14 or until an exacerbation occurred. RESULTS: After 14 d, subjects consuming the low-antioxidant diet had a lower percentage predicted forced expiratory volume in 1 s and percentage predicted forced vital capacity than did those consuming the high-antioxidant diet. Subjects in the low-antioxidant diet group had increased plasma C-reactive protein at week 14. At the end of the trial, time to exacerbation was greater in the high-antioxidant than in the low-antioxidant diet group, and the low-antioxidant diet group was 2.26 (95% CI: 1.04, 4.91; $P = 0.039$) times as likely to exacerbate. Of the subjects in the low-antioxidant diet group, no difference in airway or systemic inflammation or clinical outcomes was observed between the groups that consumed the tomato extract and those who consumed placebo. CONCLUSIONS: Modifying the dietary intake of carotenoids alters clinical asthma outcomes. Improvements were evident only after increased fruit and vegetable intake, which suggests that whole-food interventions are most effective. This trial was registered at http://www.actr.org.au as ACTRN012606000286549.